

### Question 1

Which statements about attention mechanism are true?

- ☒ a. The scaled dot-product attention mechanism generally has the form  $\text{softmax}(\frac{1}{\sqrt{d}} QK^T)V$ , where  $K, Q, V$  are keys, queries, and values, respectively, and  $d$  is the first dimension of both  $K$  and  $Q$ .
  - b. A spatial attention mechanism causes the network to treat each pixel in an image as equally important. Lx
  - ☒ c. An LSTM with self-attention has access to all previous hidden states and cell states via a softmax-based attention mechanism.
  - ☒ d. The LSTM gates can be interpreted as a temporal attention mechanism.
  - e. The attention score  $e(i)$  is obtained from the attention vector  $a(i)$  via the softmax function.
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### Question 2

Which statements about the gated recurrent unit (GRU, Cho et al., 2014) are true?

- a. The GRU has one gate more than the fully featured LSTM architecture.
- ☒ b. A forget gate mechanism is the core of the GRU architecture. L loss
- ☒ c. In the GRU, a so-called "reset gate"  $r(t)$  regulates how strongly the increments  $z(t)$  depend on the hidden  $h(t)$ .
- d. Usually, the GRU has more parameters than an LSTM with the same number of hidden units.
- ☒ e. The so-called "update gate" L loss of the GRU acts as a coupled input and forget gate.

### Question 3

Which statements about LSTM applications are true?

- ☒ a. LSTM can be used for protein classification. Hereby, the protein is represented as a sequence of amino acids and the labels are either functional or structural properties of the respective protein.
- b. "word2vec" (Mikolov et al., 2013) is an example for a sentence embedding technique. Lx
- ☒ c. A word embedding space is a real vector space in which words are represented as vectors (points) which often show meaningful relations to each other. For instance, let  $f(\cdot)$  be the (trained) embedding function, then  $f(king) - f(man) + f(woman)$  is found close to the point  $f(queen)$ .
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☒ d. Radford et al. (2017) used an LSTM to discover sentiment in Amazon reviews by training it for sequence classification (good vs. bad) on the review text. next character
- ☒ e. Fan et al. (2015) trained an LSTM to predict facial features from audio/text inputs. On this basis, given a speech recording, they could generate photo-realistic videos of talking heads.

#### Question 4

Which of the following statements about LSTM applications are true?

- a. Karpathy and Fei-Fei (2015) propose to perform image captioning using a CNN image encoder network in combination with an LSTM next-character text generation network.
- ☒ b. Alex Graves showed in his 2013 paper how LSTM can be used to generate text sequences simply by predicting one character at a time.
- ☒ c. Sequence-to-sequence learning tasks without one-to-one correspondences can be tackled using an encoder-decoder architecture.
- d. Language translation is an example for sequence-to-sequence learning tasks with one-to-one correspondences.
- ☒ e. Part-of-speech tagging is an example for sequence-to-sequence learning with one-to-one correspondences.
- ☒ f. Hochreiter et al. (2001) showed how RNNs can "learn to learn" by arranging labelled datasets as sequences. At time  $t$ , the RNN takes an input  $x_t$  along with the label  $y_t$  and makes a prediction  $\hat{y}_t$ .

#### Question 5

Which statements about Transformer networks (Vaswani et al., 2017) are true?

- ☒ a. Without positional embeddings, the Transformer network has no way of figuring out the order of elements in an input sequence.
- b. The Transformer architecture consists of an encoder and a decoder part, where the encoder depends on the output of the decoder. *other way round*
- ☒ c. Transformer networks use multi-head attention, which works as follows. We run multiple instances of scaled dot-product attention, concatenate their outputs (or context) and feed the result into a linear layer.
- d. The Transformer architecture suffers from the vanishing gradient problem in the time dimension.
- e. BERT (Devlin et al., 2019) uses two different pre-training objectives, (1) inserting missing (masked out) words into a sentence, and (2) next sentence prediction, i.e., generating a response sentence to a question sentence.

predict if successor of sentence

### Question 6

Which statements about LSTM variants are true?

- ☒ a. Peephole connections (Gers and Schmidhuber, 2000) provide gates with direct read access to the cell state.
- ☒ b. A Bidirectional LSTM (Schuster et al., 1997) has advantages over the unidirectional version if the input sequences do not have a preferred processing direction.
- c. The multidimensional LSTM (Graves et al., 2007) can process sequences consisting of vectors which stem from high-dimensional vector spaces.
- d. Stacked LSTM (Graves et al., 2013) stack LSTM layer by using one LSTM's final cell state as the initial cell state for the next LSTM.
- ☒ e. Convolutional LSTM (Shi et al., 2015) can process sequences of images, e.g., videos.

### Question 7

Which statements about LSTM dropout are correct?

- a. LSTM needs special dropout techniques because a naive application of dropout will result in vanishing/exploding gradients.
- b. It is especially important to apply dropout when evaluating the model.
- ☒ c. LSTM needs special dropout techniques because a naive application will reset any memory cell with exponentially high probability.
- ☒ d. Dropout is a regularization technique that randomly sets activations to zero during the forward pass.
- ☒ e. The central idea of zoneout is to randomly clear the updates to the memory cell instead of its content.